

CSA-1610

COI - AT1

S. Md. Irfan

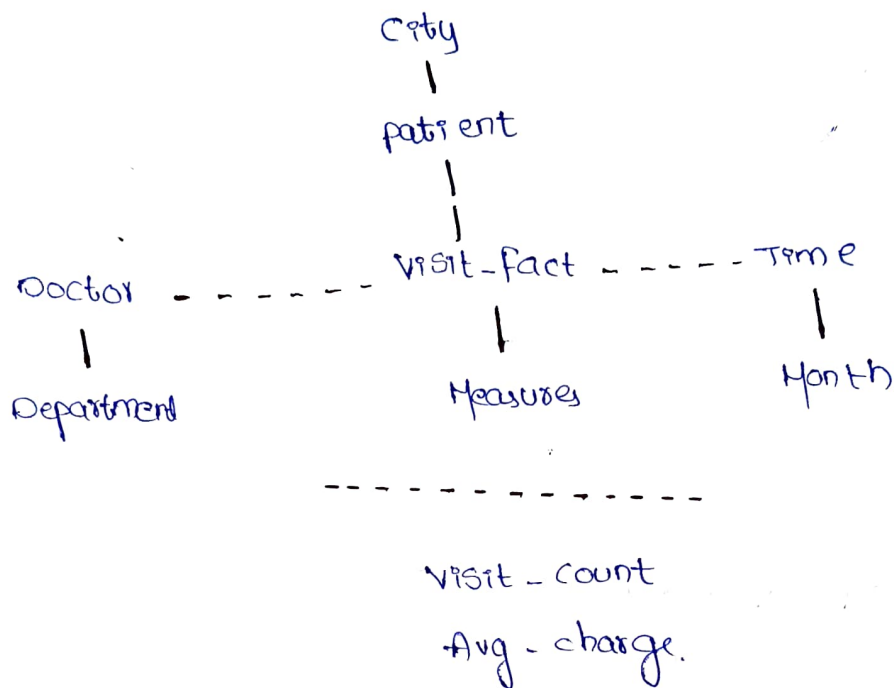
192373001

C.S.E (D.S.)

Q1)
sol

City Hospital Data warehouse

(a) Snowflake Schema.



Fact Table : visit-fact

- * patient - ID
- * Doctor - ID
- * Department - ID
- * Time - ID
- * visit-count
- * Avg-charge.

Dimension Table:

patients

- * patient - ID
- * Name

* Age

* Gender

* City

Doctor

* Doctor - ID

* Name

* Specialization

Department

Department - ID

Department - Name

Time

* Date

* Month

* Quarter

* Year

b) OLAP Operations

starting cuboid : [patient, Doctor, Department, Time]

Operations:

1. Roll-up Patient $\rightarrow$ ALL
2. Roll-up Doctor $\rightarrow$ ALL
3. Roll-up Time $\rightarrow$ Year
4. Keep department unchanged
5. Calculate Avg (Avg-change)

Final Cuboid :

[ALL, ALL, Department, Year]

c) Number of cuboids

Dimensions = 4

level per dimension = 4

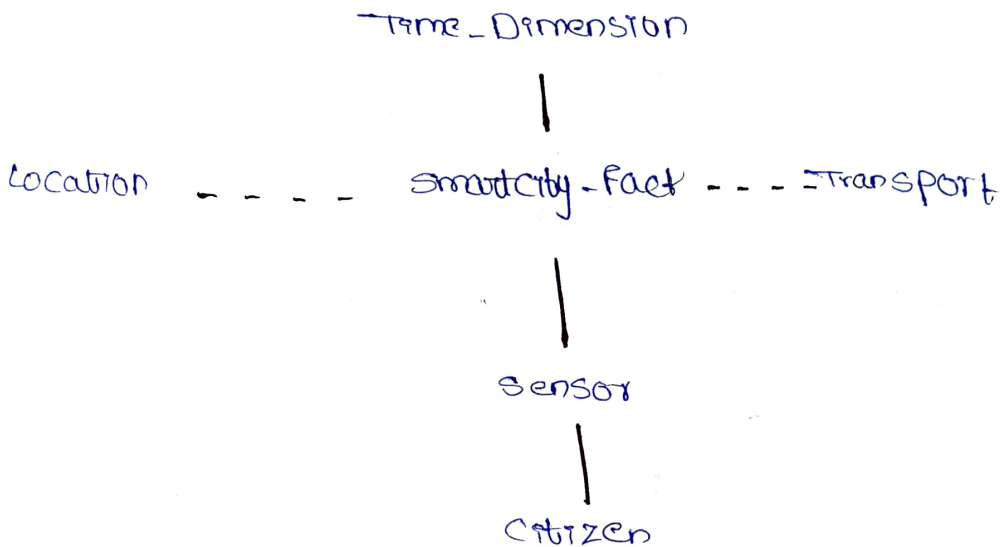
Formula $\Rightarrow$ cuboids = $4 \times 4 \times 4 \times 4 = 4^4 = 256$ cuboids.

Q2)

sol

Small City Star Schema.

Star schema.



Fact Table

Smart-city Fact

Measures :

- Vehicle count
- Congestion level
- Energy consumption
- Air quality index
- Noise level
- Ridership
- Punctuality
- Citizen feedback score.

Dimension Tables

Time

- * Day
- * Month
- * Year

Location

- * City
- * Zone
- * Road

Sensor

- * Sensor ID
- * Sensor Type

Transport

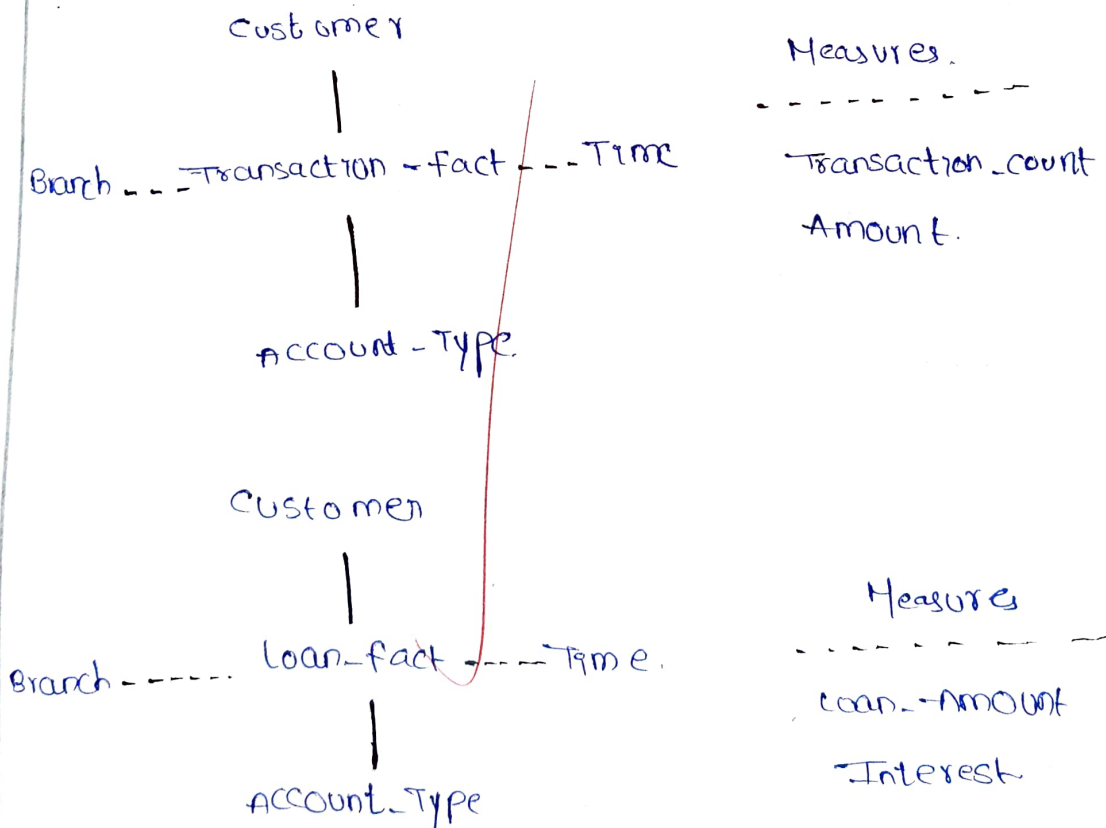
- * Bus
- * Metro
- * Route

Citizen

- * User ID
- * Age group
- * Feedback category

Q3) Banking Data Warehouse.

a) Galaxy Schema.



Shared Dimensions

- customer
- Branch
- Time
- Account type.

b) OLAP operations

Base cuboid : [customer, Branch, Time - Account - Type]

operations

- * Roll-up customer → ALL
- * Roll-up Account-Type → ALL
- * Roll-up Time → year
- * Keep Branch

Measures

- Avg(Amount)
- Avg(loan-Amount)

→ final cuboid : [ALL, Branch, year, ALL]

c) Cuboids

Dimension = 4 , level = 4

Total cuboids = $4^4 = 256$ cuboids

Q4) Manufacturing company

Roll up : Summarize defect rates by product category for the current year across all factories

operation

- product → category
- factory → ALL
- time → current year

Measure

- * Defect rate

Slice

condition : supplier rating ≥ 4

Measure : Production units

Across : all regions

Drill down

- * factory → Department
- * product → product type

Measure

Defect rate

Drce

- * filters : selected product
- * selected factory location

Measures :

- * Defect rate
- * Supplier reliability.

Q5) Retail chain

Roll up

Summarize Inventory Turnover by product category.

Drill Down

Filter :

- * product category = Electronics
- * Region = Selected region

Drill down to

- * Sub category
- * SKU

Measures

- * Sales revenue
- * Units sold

Slice

Filter

Gender = Male

Membership = Bronze

Measures : Sales revenue

Across : All stores

Dice

Filters

- * Selected cities
- * Selected product categories.
- * Selected time periods.

Measures

- sales revenue
- units sold.